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MODEL SELECTION AND VALIDATION REPORT ( other_fd_biomass model )
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## OBSERVATIONS:3390

## SANITY: OK − all models in the selection process passed sanity checks.

## SELECTED RANDOM TERM: spatiotemporal_random_fields_ar1

## MODEL SUMMARY ( other_fd_biomass model )

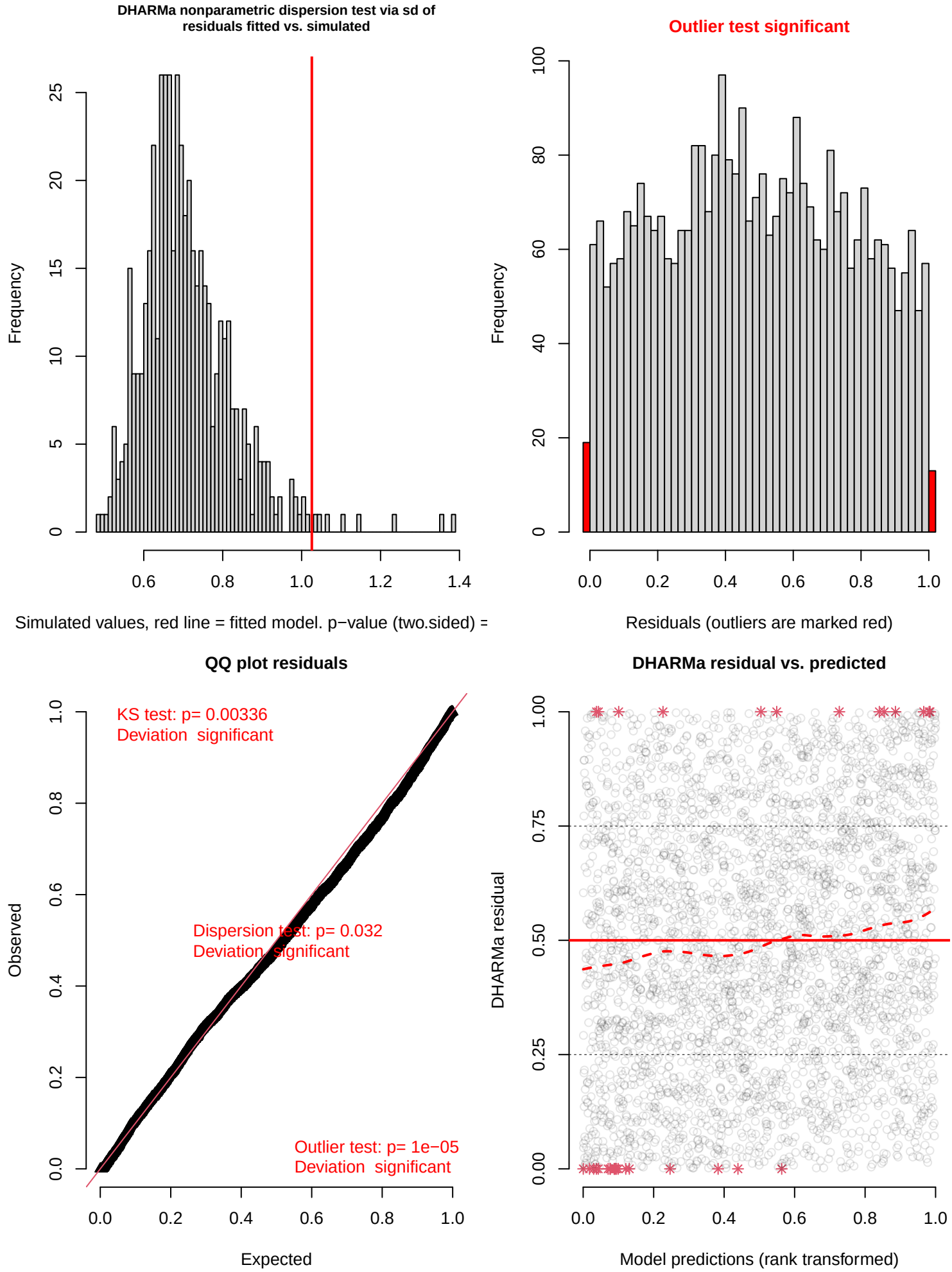
Spatiotemporal model fit by REML ['sdmTMB']
Formula: fixed_formula
Time column: year
Family: lognormal(link = 'log')

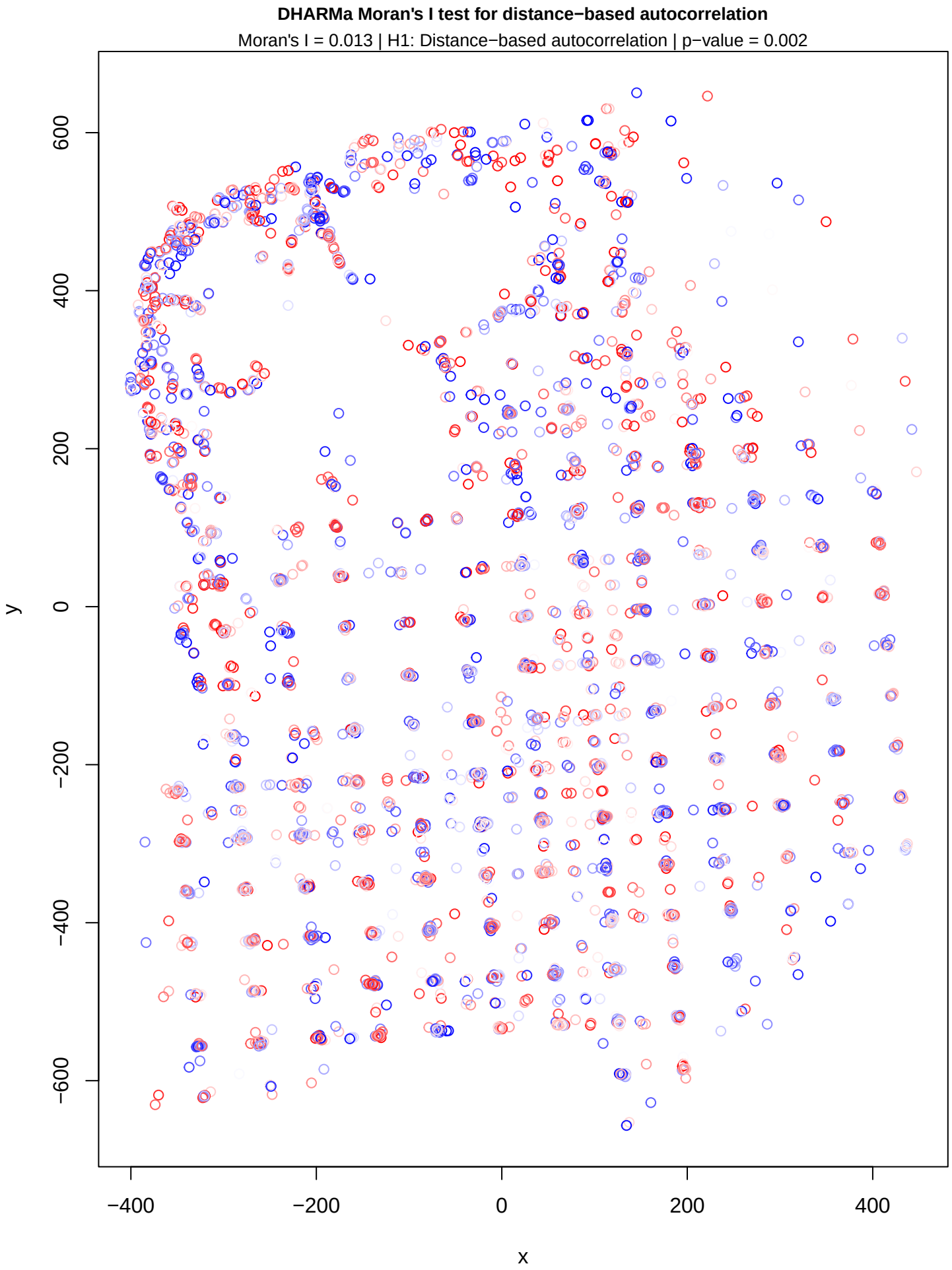
Conditional model:
              coef.est coef.se
(Intercept)       7.01    0.17
ch_fric           0.08    0.02
ch_feve          −0.16    0.01
tric              0.01    0.02
teve             −0.32    0.02
sbt_mean          0.30    0.05
sic_mean          0.05    0.05
sbt_sd            0.12    0.03
sst_sd            0.05    0.03
log_chla_mean     0.03    0.02
depth            −0.04    0.03

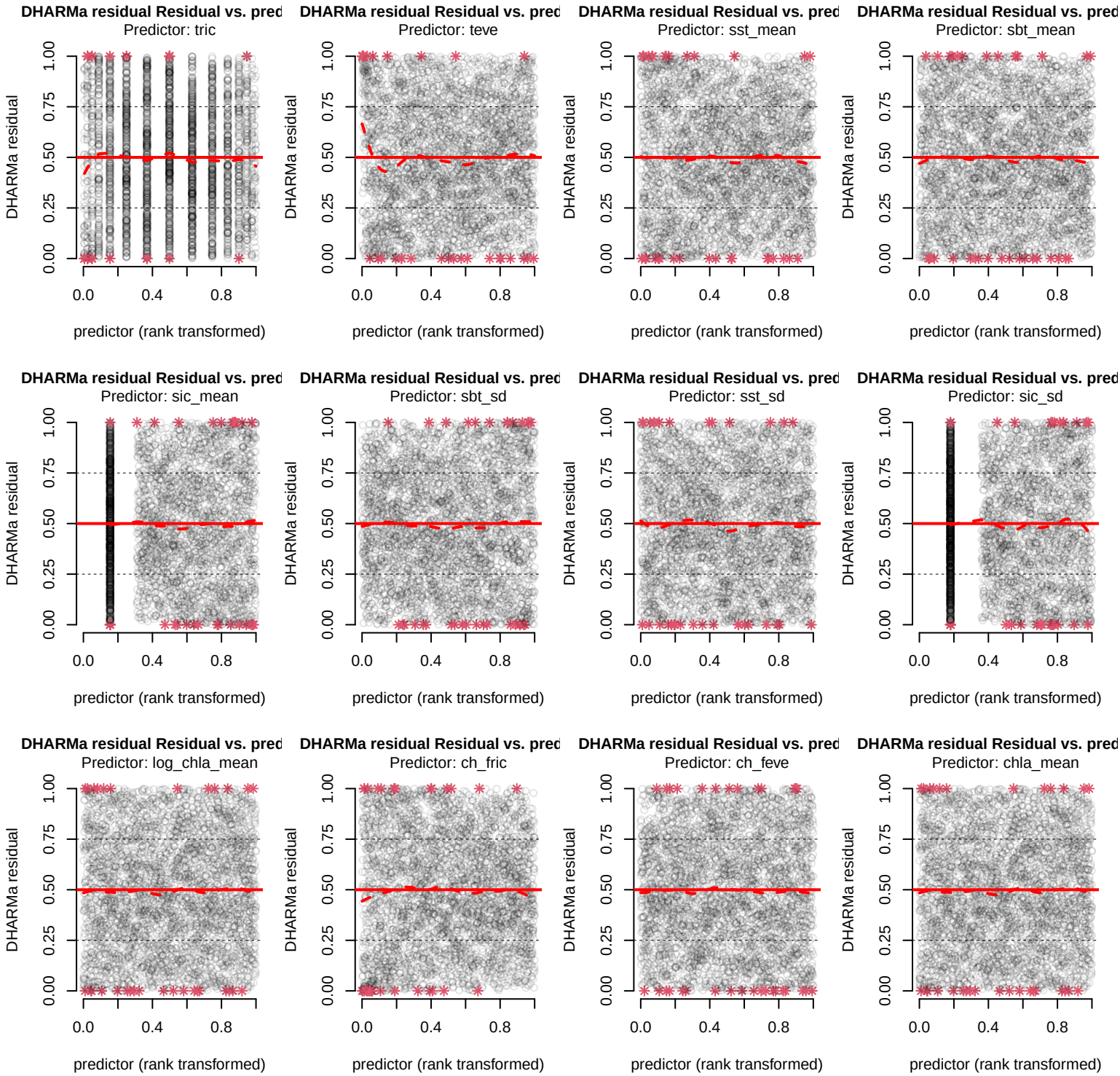
Dispersion parameter: 0.74
Spatiotemporal AR1 correlation (rho): 0.86
Matérn range: 296.39
Spatiotemporal marginal AR1 SD: 0.79
REML criterion at convergence: 27058.216

See ?tidy.sdmTMB to extract these values as a data frame.

## SELECTION STATISTICS
                                                   df      AIC  delta_AIC
spatiotemporal_random_fields_ar1                   15 54146.43   2.355853
spatiotemporal_random_fields_ar1_with_anisotropy 17 54148.79 259.766713
spatiotemporal_random_fields_iid                   14 54408.55   1.592338
spatiotemporal_random_fields_iid_with_anisotropy 16 54410.15 271.013345
spatial_random_fields                              14 54681.16   3.187558
spatial_random_fields_with_anisotropy              16 54684.35 569.995345
error_only                                         12 55254.34         NA
```


DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated
Frequency
Simulated values, red line = fitted model. p−value (two.sided) =
Outlier test significant
Frequency
Residuals (outliers are marked red)
QQ plot residuals
KS test: p= 0.00336
Deviation significant
Dispersion test: p= 0.032
Deviation significant
Outlier test: p= 1e−05
Deviation significant
Observed
Expected
DHARMa residual vs. predicted
DHARMa residual
Model predictions (rank transformed)


DHARMa Moran's I test for distance−based autocorrelation
Moran's I = 0.013 | H1: Distance−based autocorrelation | p−value = 0.002
y
600
400
200
0
−200
−400
−600
x
−400
−200
0
200
400


DHARMa residual Residual vs. pred
Predictor: tric
Predictor: teve
Predictor: sst_mean
Predictor: sbt_mean
Predictor: sic_mean
Predictor: sbt_sd
Predictor: sst_sd
Predictor: sic_sd
Predictor: log_chla_mean
Predictor: ch_fric
Predictor: ch_feve
Predictor: chla_mean
DHARMa residual
0.00
0.25
0.50
0.75
1.00
0.0
0.4
0.8
predictor (rank transformed)